

Yue Chu

PhD, MSPH, MBBS

Department of Sociology, Institute for Population Research
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RESEARCH AREAS

Population health, demography, epidemiology, global health estimates, mortality modeling, demographic and disease surveillance, AI in global health, data systems, health disparities.

EDUCATION AND TRAINING

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| 2025 | PhD in Sociology , The Ohio State University, Columbus, USA
Minor in Statistics and Demography
<i>Dissertation: Leveraging Language Models and Machine Learning in Verbal Autopsy Analysis</i> |
| 2020 | MA in Sociology , The Ohio State University, Columbus, USA
<i>Thesis: SVD-Bayes: A Singular Value Decomposition-Based Approach Under Bayesian Framework for Indirect Estimation of Age-Specific Fertility and Mortality</i> |
| 2014 | MSPH , Johns Hopkins Bloomberg School of Public Health, Baltimore, USA
Concentration in Population, Family and Reproductive Health
<i>Thesis: The influence of family environment factors on children's consumption of healthy foods among 5-9 years old children from low-income families in New York City</i> |
| 2012 | Bachelor of Medicine , Fudan University, Shanghai, China
Specialty in Preventive Medicine
<i>Thesis: Evaluation of tuberculosis patient management strategies: evidence from a randomized controlled trial</i> |

PROFESSIONAL EXPERIENCE

Feb 2026–present	Postdoctoral Fellow , Institute for Population Research, The Ohio State University, Columbus, OH
May 2025–Jan 2026	Postdoctoral Fellow , Health and Environment Modeling Laboratory, The Ohio State University, Columbus, OH
Aug 2023–Sep 2023	Visiting Scholar , Africa Health Research Institute, Somkele, South Africa
May 2022–May 2023	Data Science Intern , Nationwide Mutual Insurance Company / Nationwide Center for Advanced Customer Insights, Columbus, OH
Jan 2019–May 2025	Graduate Research Associate , Institute for Population Research, The Ohio State University, Columbus, OH
Jan 2015–Jul 2018	Research Associate , Department of International Health, Johns Hopkins Bloomberg School of Public Health, Baltimore, MD
Sep 2014–Jan 2015	Research Assistant , Department of Health Policy and Management, Johns Hopkins Bloomberg School of Public Health
Apr 2013–Sep 2014	Research Assistant , Johns Hopkins Bloomberg School of Public Health
May 2013–Aug 2013	Research Assistant , Technical Division, United Nations Population Fund (UNFPA), New York
Apr 2012–Jun 2012	Research Assistant , Department of Health Statistics and Social Medicine, School of Public Health, Fudan University, Shanghai, China
Feb 2012–Mar 2012	Intern , Division of Vital Statistics, Center for Disease Control and Prevention, Shanghai, China
Mar 2011–Aug 2011	Clinical Intern , Fifth People's Hospital, Shanghai, China

PUBLICATIONS

Papers

1. **Chu, Y.** Leveraging language models and machine learning in verbal autopsy analysis [preprint]. *arXiv*. 2025. <https://doi.org/10.48550/arXiv.2508.19274>
2. **Chu, Y.**, Marston, M., Dube, A., Festo, C., Geubbels, E., Gregson, S., Herbst, K., et al. Temporal changes in cause of death among adolescents and adults in six countries in eastern and southern Africa in 1995-2019: A multi-country surveillance study of verbal autopsy data. *Lancet Global Health*. 2024;12(8):e1278-e1287. [https://doi.org/10.1016/S2214-109X\(24\)00171-2](https://doi.org/10.1016/S2214-109X(24)00171-2)
3. Damerow, S. M., Yeung, D., Martins, J. S., Pathak, I., **Chu, Y.**, Liu, L., Fisker, A. B. Provider-mother interactions are associated with birth outcome misclassifications in household surveys: A case-control study in Guinea-Bissau. *Journal of Global Health*. 2023;13:04086. <https://doi.org/10.7189/jogh.13.04086>
4. Liu, Y., **Chu, Y.**, Yeung, D., Wang, W., Wang, L., Yin, P., Liu, J., Zhou, M., Liu, L. National and sub-national levels and causes of mortality among 5-19-year-olds in China in 2004-2019: A systematic analysis of evidence from the Disease Surveillance Points System. *Journal of Global Health*. 2022;12:11008. <https://doi.org/10.7189/jogh.12.11008> [joint first authorship]
5. Perin, J., **Chu, Y.**, Villavicencio, F., Schumacher, A., McCormick, T., Guillot, M., Liu, L. Adapting and validating the log quadratic model to derive under-five age- and cause-specific mortality (U5ACSM): a preliminary analysis. *Population Health Metrics*. 2022;20(1):3. <https://doi.org/10.1186/s12963-021-00277-w>
6. Turner, A. N., Kline, D., Norris, A., Phillips, W. G., Root, E., Wakefield, J., Li, Z., Lemeshow, S., Spahnie, M., Luff, A., **Chu, Y.**, et al. Prevalence of current and past COVID-19 in Ohio adults. *Annals of Epidemiology*. 2022;67:50-60. <https://doi.org/10.1016/j.annepidem.2021.11.009>
7. Schumacher, A. E., McCormick, T. H., Wakefield, J., **Chu, Y.**, Perin, J., Villavicencio, F., Simon, N., Liu, L. A flexible Bayesian framework to estimate age- and cause-specific child mortality over time from sample registration data. *Annals of Applied Statistics*. 2022;16(1):124-143. <https://doi.org/10.1214/21-AOAS1489>
8. Kline, D., Li, Z., **Chu, Y.**, Wakefield, J., Miller, W. C., Turner, A. N., Clark, S. J. Estimating seroprevalence of SARS-CoV-2 in Ohio: A Bayesian multilevel poststratification approach with multiple diagnostic tests. *Proceedings of the National Academy of Sciences of the United States of America*. 2021;118(26):e2023947118. <https://doi.org/10.1073/pnas.2023947118>
9. Chandramohan, D., Fottrell, E., Leita, J., Nichols, E., Clark, S. J., Alsokhn, C., Cobos Munoz, D., AbouZahr, C., Di Pasquale, A., Mswia, R., Choi, E., **Chu, Y.**, et al. Estimating causes of death where there is no medical certification: Evolution and state of the art of verbal autopsy. *Global Health Action*. 2021;14(sup1):1982486. <https://doi.org/10.1080/16549716.2021.1982486>
10. Helleringer, S., Liu, L., **Chu, Y.**, Rodrigues, A., Fisker, A. B. Biases in survey estimates of neonatal mortality: Results from a validation study in urban areas of Guinea-Bissau. *Demography*. 2020;57(5):1705-1726. <https://doi.org/10.1007/s13524-020-00911-6>
11. Wahl, B., Deloria Knoll, M., Shet, A., Gupta, M., Kumar, R., Liu, L., **Chu, Y.**, et al. National, regional, and state-level pneumonia and severe pneumonia morbidity in children in India:

- modelled estimates for 2000 and 2015. *Lancet Child and Adolescent Health*. 2020;4(9):678-687. [https://doi.org/10.1016/S2352-4642\(20\)30129-2](https://doi.org/10.1016/S2352-4642(20)30129-2)
12. **Chu, Y.** SVD-Bayes: A singular value decomposition-based approach under Bayesian framework for indirect estimation of age-specific fertility and mortality [master's thesis]. The Ohio State University; 2020. https://rave.ohiolink.edu/etdc/view?acc_num=osu1609638415015896
 13. McAllister, D. A., Liu, L., Shi, T., **Chu, Y.**, Reed, C., Burrows, J., Adeloye, D., et al. Global, regional, and national estimates of pneumonia morbidity and mortality in children younger than 5 years between 2000 and 2015: a systematic analysis. *Lancet Global Health*. 2019;7(1):e47-e57. [https://doi.org/10.1016/S2214-109X\(18\)30408-X](https://doi.org/10.1016/S2214-109X(18)30408-X)
 14. Liu, L., **Chu, Y.**, Oza, S., Hogan, D., Perin, J., Bassani, D. G., Ram, U., et al. National, regional, and state-level all-cause and cause-specific under-5 mortality in India in 2000-15: a systematic analysis with implications for the Sustainable Development Goals. *Lancet Global Health*. 2019;7(6):e721-e734. [https://doi.org/10.1016/S2214-109X\(19\)30080-4](https://doi.org/10.1016/S2214-109X(19)30080-4) [joint first authorship]
 15. Wahl, B., Sharan, A., Deloria Knoll, M., Kumar, R., Liu, L., **Chu, Y.**, McAllister, D. A., et al. National, regional, and state-level burden of *Streptococcus pneumoniae* and *Haemophilus influenzae* type b disease in children in India: modelled estimates for 2000-15. *Lancet Global Health*. 2019;7(6):e735-e747. [https://doi.org/10.1016/S2214-109X\(19\)30081-6](https://doi.org/10.1016/S2214-109X(19)30081-6)
 16. Choudhary, T. S., Sinha, B., Khera, A., Bhandari, N., **Chu, Y.**, Jackson, B., Walker, N., et al. Factors associated with the decline in under-five diarrhea mortality in India: a LiST analysis. *Journal of Global Health*. 2019;9(2):020804. <https://doi.org/10.7189/jogh.09.020804>
 17. Black, R., Fontaine, O., Lamberti, L., Bhan, M., Huicho, L., El Arifeen, S., Masanja, H., **Chu, Y.**, et al. Drivers of the reduction in childhood diarrhea mortality 1980-2015 and interventions to eliminate preventable diarrhea deaths by 2030. *Journal of Global Health*. 2019;9(2):020801. <https://doi.org/10.7189/jogh.09.020801>
 18. Countdown to 2030 Collaboration. Countdown to 2030: Tracking progress towards universal coverage for reproductive, maternal, newborn and child health. *Lancet*. 2018;391(10129):1538-1548. [https://doi.org/10.1016/S0140-6736\(18\)30104-1](https://doi.org/10.1016/S0140-6736(18)30104-1)
 19. Wahl, B., O'Brien, K. L., Greenbaum, A., Majumder, A., Liu, L., **Chu, Y.**, Luksic, I., et al. Burden of *Streptococcus pneumoniae* and *Haemophilus influenzae* type b disease in children in the era of conjugate vaccines: global, regional, and national estimates for 2000-15. *Lancet Global Health*. 2018;6(7):e744-e757. [https://doi.org/10.1016/S2214-109X\(18\)30247-X](https://doi.org/10.1016/S2214-109X(18)30247-X)
 20. He, C., Liu, L., **Chu, Y.**, Perin, J., Dai, L., Li, X., Miao, L., et al. National and subnational all-cause and cause-specific child mortality in China, 1996-2015: a systematic analysis with implications for the Sustainable Development Goals. *Lancet Global Health*. 2017;5(2):e186-e197. [https://doi.org/10.1016/S2214-109X\(16\)30334-5](https://doi.org/10.1016/S2214-109X(16)30334-5)
 21. Wilson, L. M., Rebholz, C. M., Jirru, E., Liu, M. C., Zhang, A., Gayleard, J., **Chu, Y.**, Robinson, K. A. Benefits and harms of osteoporosis medications in patients with chronic kidney disease: a systematic review and meta-analysis. *Annals of Internal Medicine*. 2017;166(9):649-658. <https://doi.org/10.7326/M16-2752>
 22. Liu, L., Oza, S., Hogan, D., **Chu, Y.**, Perin, J., Zhu, J., Lawn, J. E., et al. Global, regional, and national causes of under-5 mortality in 2000-15: an updated systematic analysis with